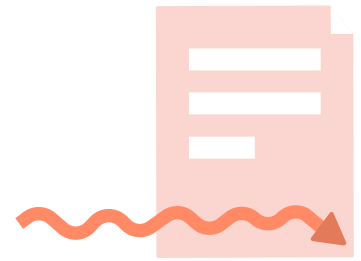


KHALIL SAHIR

M T H 6 2 2



2 MARKS

Rotating Axes Theorem statement

definition of kinetic energy, classical

3 MARKS

Problem Statement

Calculate the moment of inertia of a hoop of mass M and radius r about an axis passing its center. Let M be the mass and r the radius of the hoop. Then we can define the density of the hoop by

$$\rho = \frac{\text{mass}}{\text{area}} = \frac{M}{2\pi r dr}$$

Two objects push off against one another starting from a stationary position. First object with mass 65 kg acquires a velocity of 2 m/s to the right. What velocity does the second object having mass 70 kg acquire?

5 MARKS

Prove that the force field is non-conservative. force is given.

To find the moment of inertia of a solid circular cylinder of radius a , mass M and the height of the cylinder h about the axis of the cylinder.

$$I_{shell} = r^2 \frac{2M}{a^2} r dr = \frac{2M}{a^2} r^3 dr$$

Thus the total moment of inertia of the solid circular cylinder will be ??

all paper is from past final files...

